



Departamento de Informática

Bacharelado em Ciência da Computação

CI1212 Arquitetura de Computadores

Memória Virtual

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Memória virtual

Objetivos

Entender modelo de memória virtual e seu impacto no desempenho



Memória virtual

Motivação

Ampliar memória disponível

Proteção entre processos

Realocação



Comparação com cache

Tradução endereços

Uso da localidade de referência

Ilusão de algo, mas agora de maior capacidade em vez de velocidade

Esquema básico

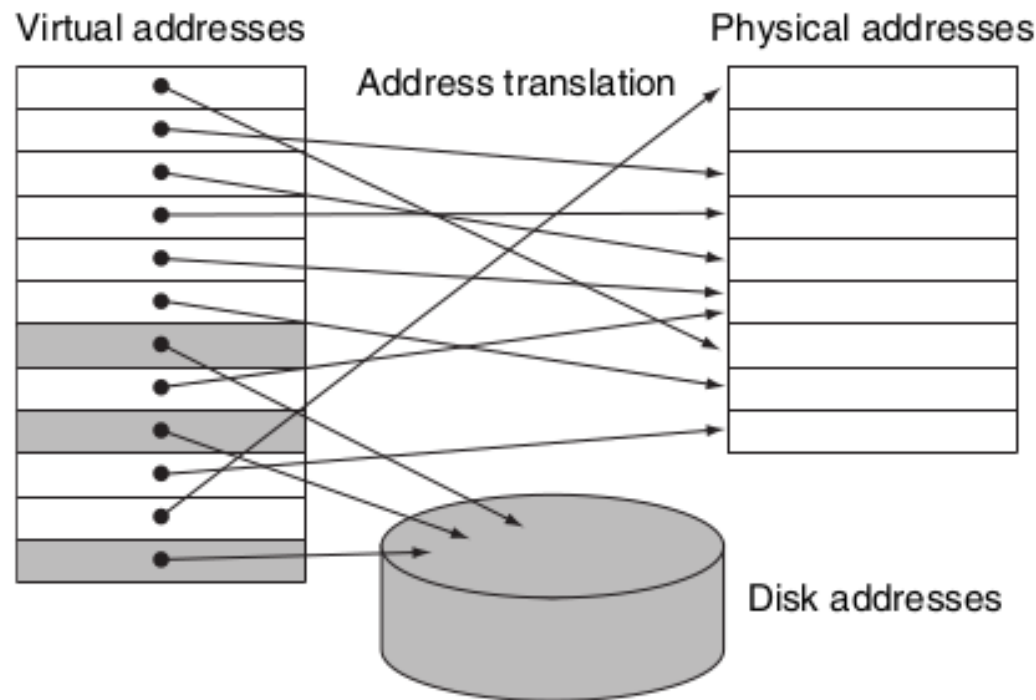


FIGURE 5.25 In virtual memory, blocks of memory (called *pages*) are mapped from one set of addresses (called *virtual addresses*) to another set (called *physical addresses*). The processor generates virtual addresses while the memory is accessed using physical addresses. Both the virtual memory and the physical memory are broken into pages, so that a virtual page is mapped to a physical page. Of course, it is also possible for a virtual page to be absent from main memory and not be mapped to a physical address; in that case, the page resides on disk. Physical pages can be shared by having two virtual addresses point to the same physical address. This capability is used to allow two different programs to share data or code.

Mapeamento de endereços

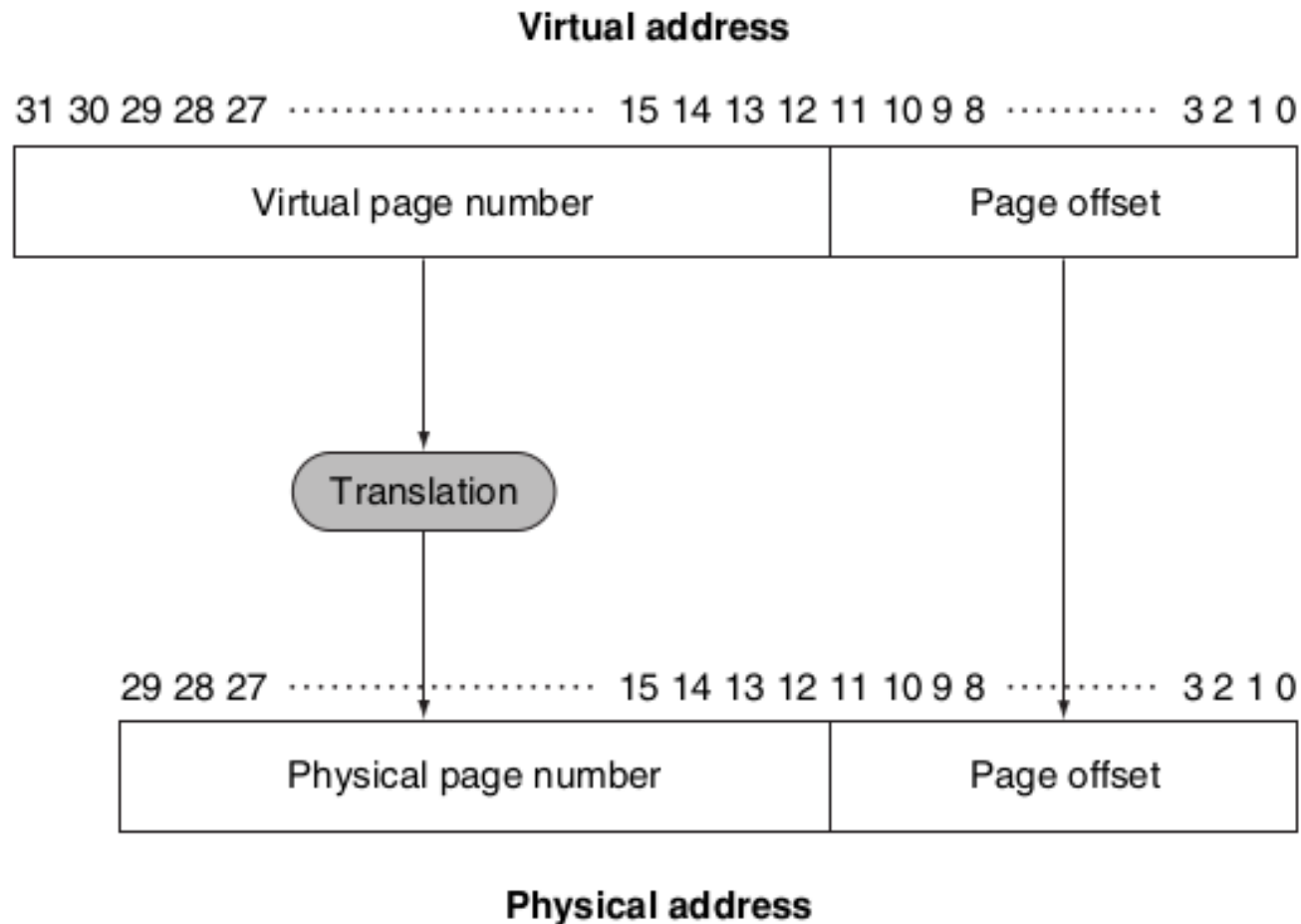


FIGURE 5.26 Mapping from a virtual to a physical address. The page size is $2^{12} = 4$ KiB. The number of physical pages allowed in memory is 2^{18} , since the physical page number has 18 bits in it. Thus, main memory can have at most 1 GiB, while the virtual address space is 4 GiB.



Decisões de projeto

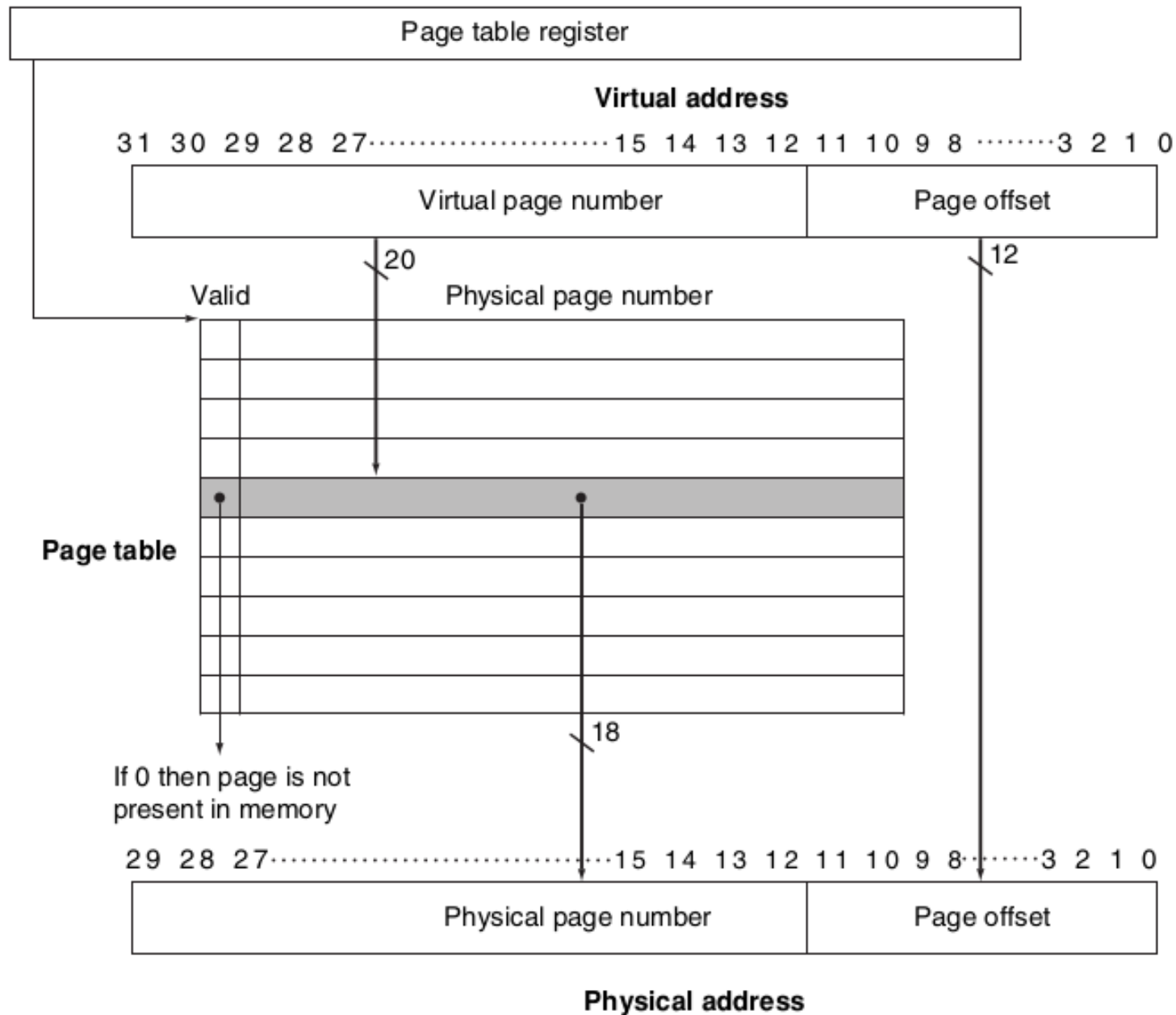
Páginas devem ser grandes para compensar grande tempo de acesso, 4KiB-16KiB, ou até mais

Reduzir page fault, como?

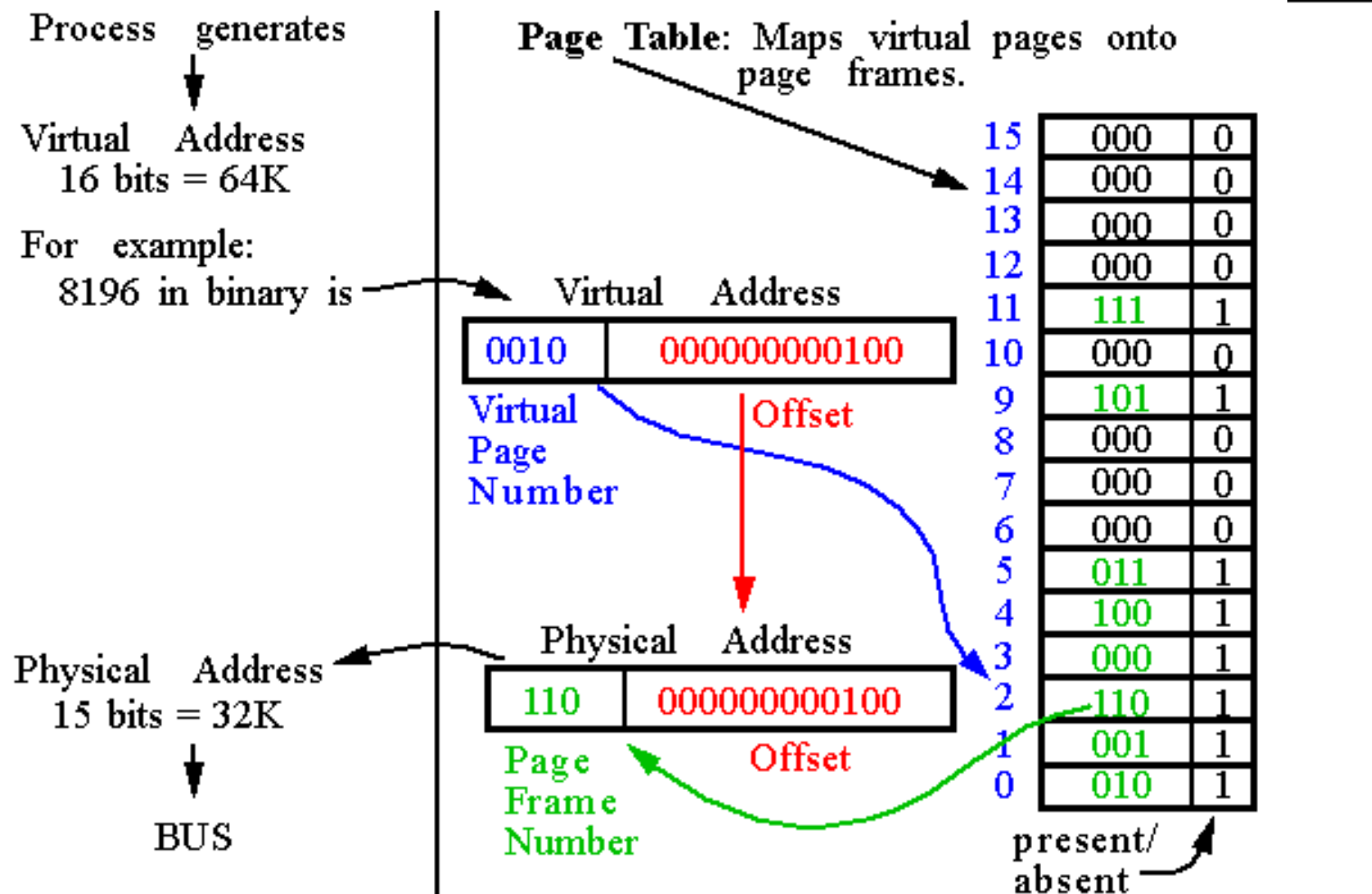
Page fault é tratado em sw, por quê?

Write-through não funciona, por quê?

Page table



Exemplo



Page table + swap info

Virtual page number

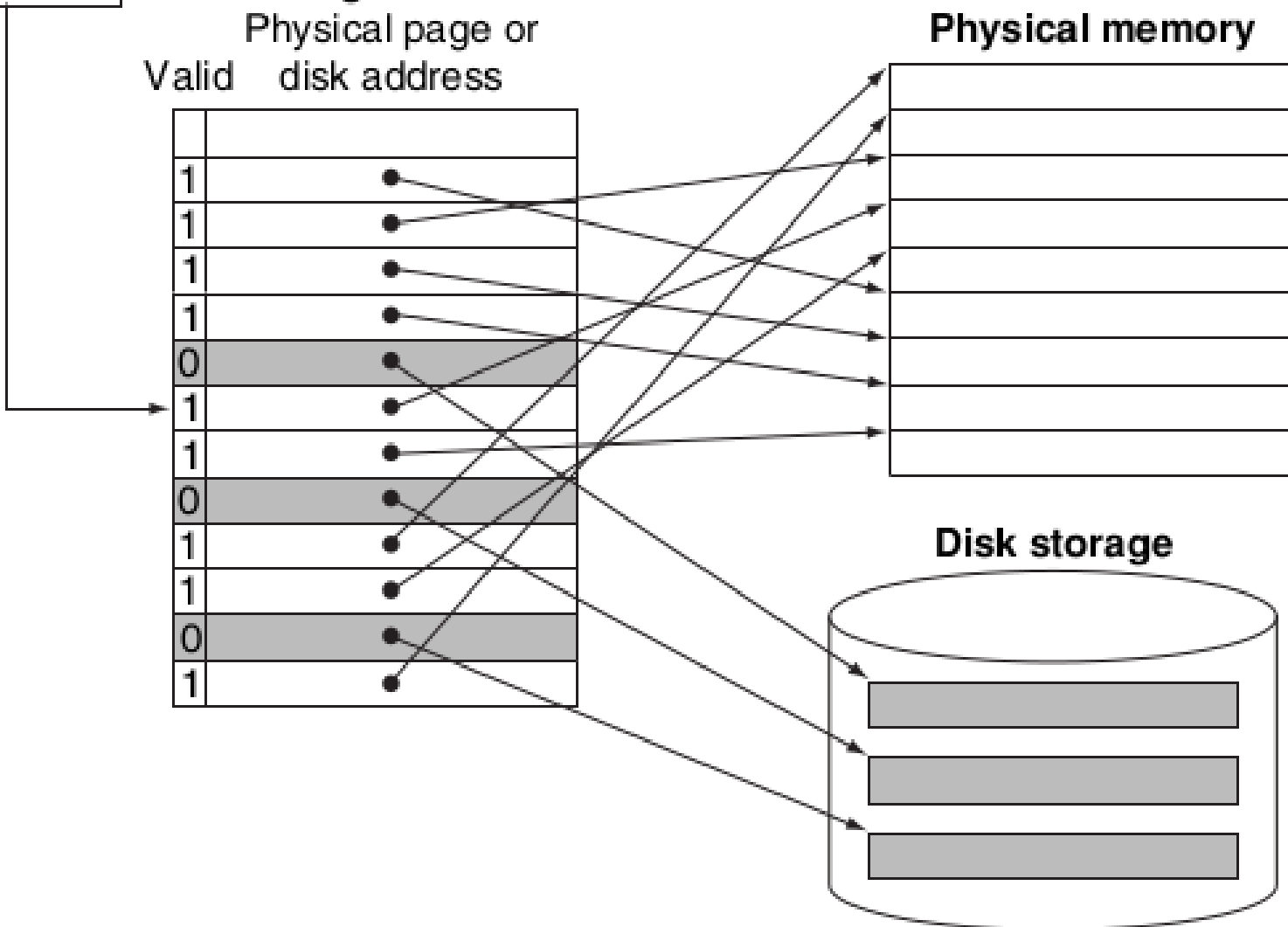
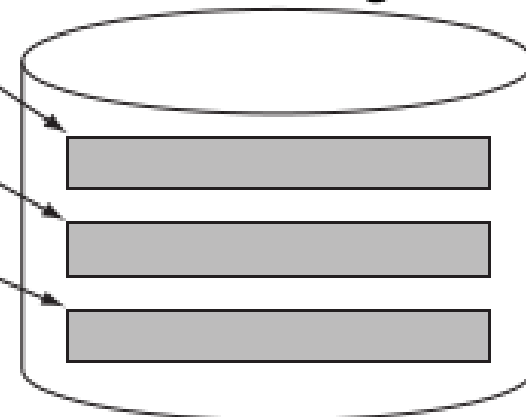
Page table

Valid Physical page or
disk address

1	•
1	•
1	•
1	•
0	•
1	•
1	•
0	•
1	•
1	•
0	•
1	•

Physical memory

Disk storage



Tamanho da page table

Pode ser enorme:

ex: 32 bits EV, páginas de 4KiB,

$$\text{entradas} = 2^{32}/2^{12} = 2^{20}$$

tamanho =

#entradas x tamanho de cada entrada



Tentar reduzir tamanho

Limitar tamanho, expandindo se necessário

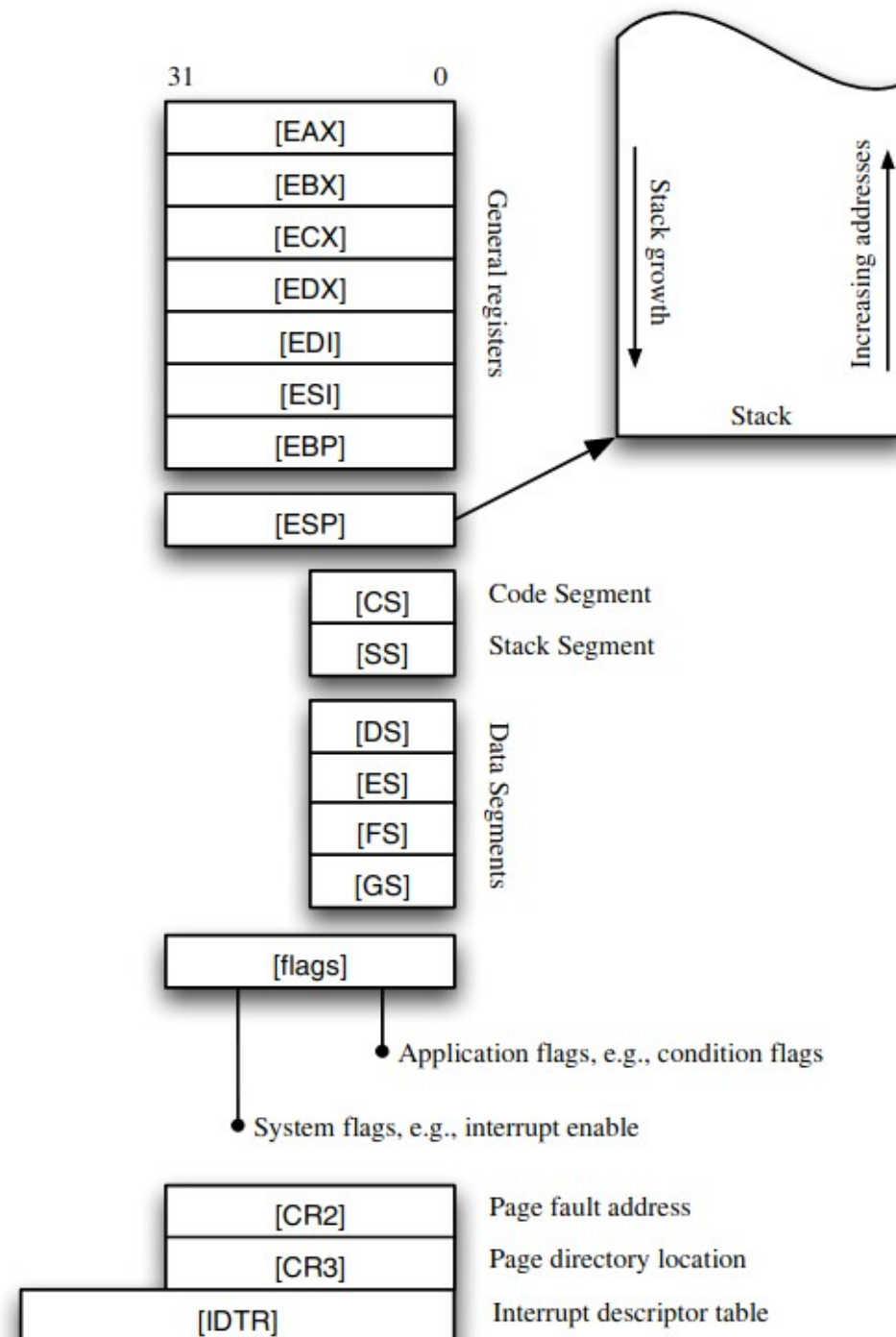
Separar áreas para stack e heap

Hashing function

Esquema hierárquico

Paged table pages :)

IA-32



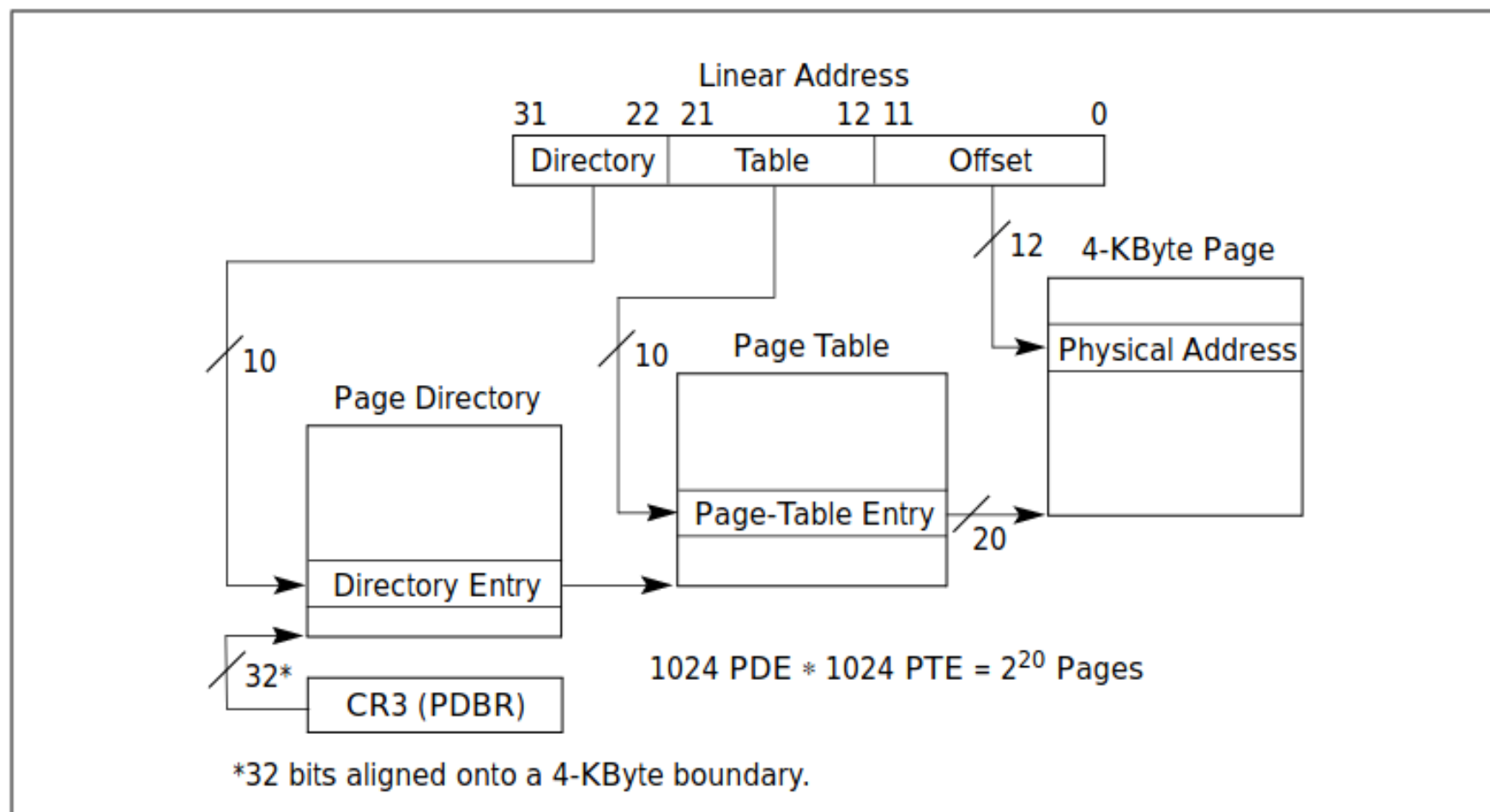


Figure 3-12. Linear Address Translation (4-KByte Pages)

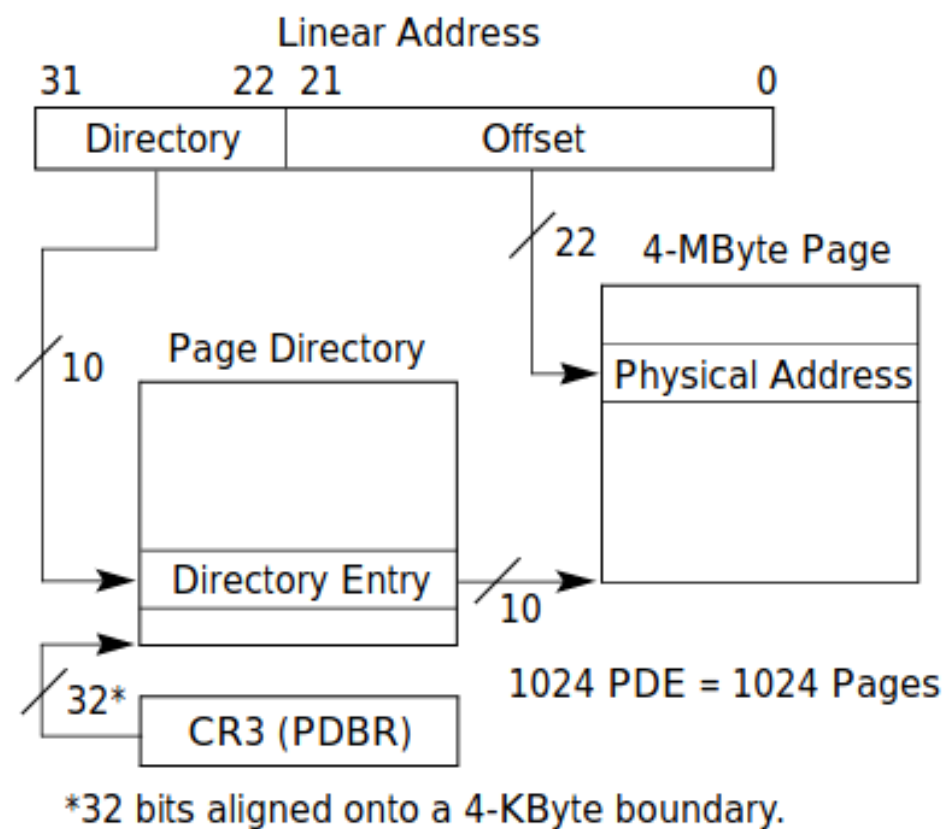
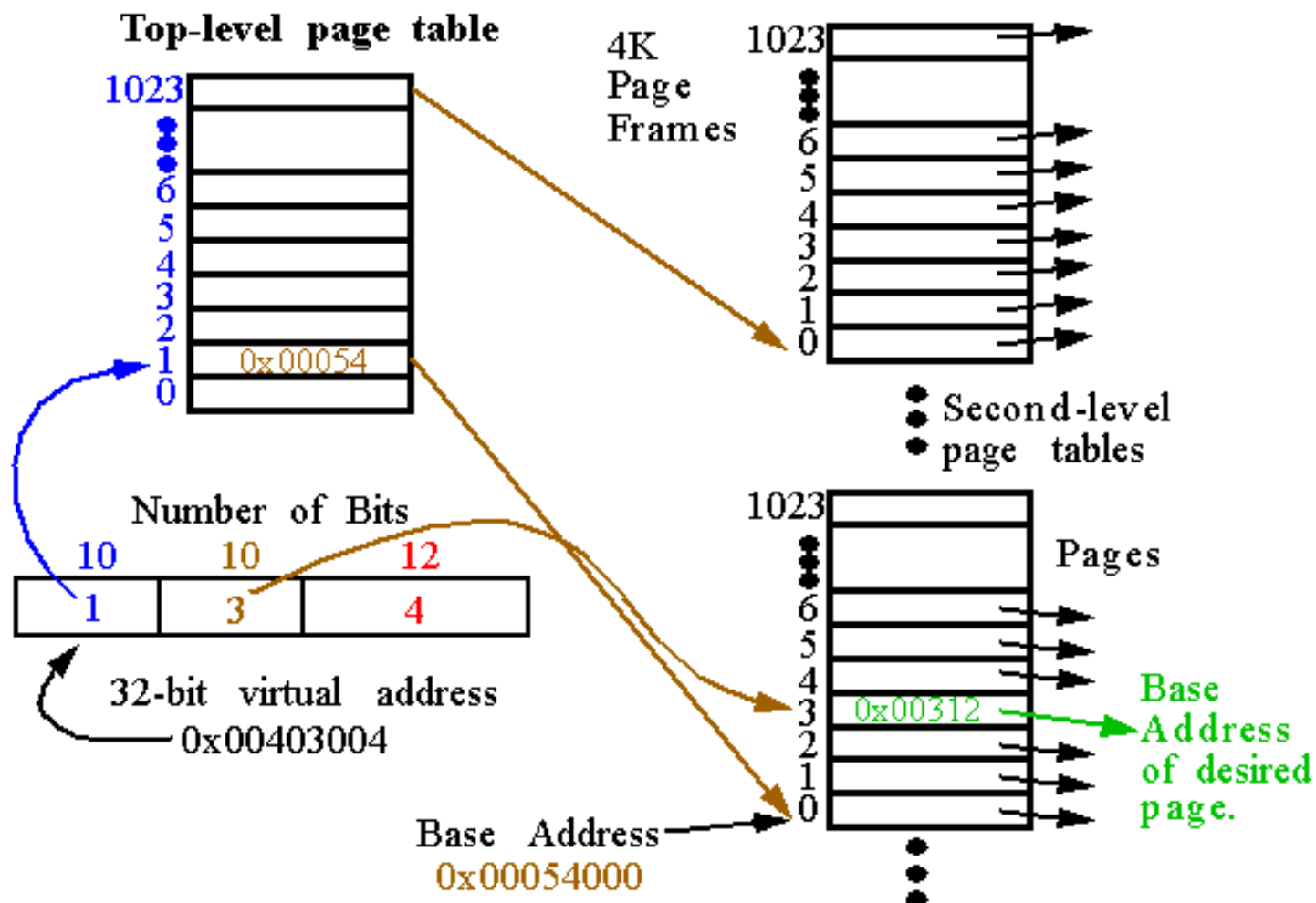
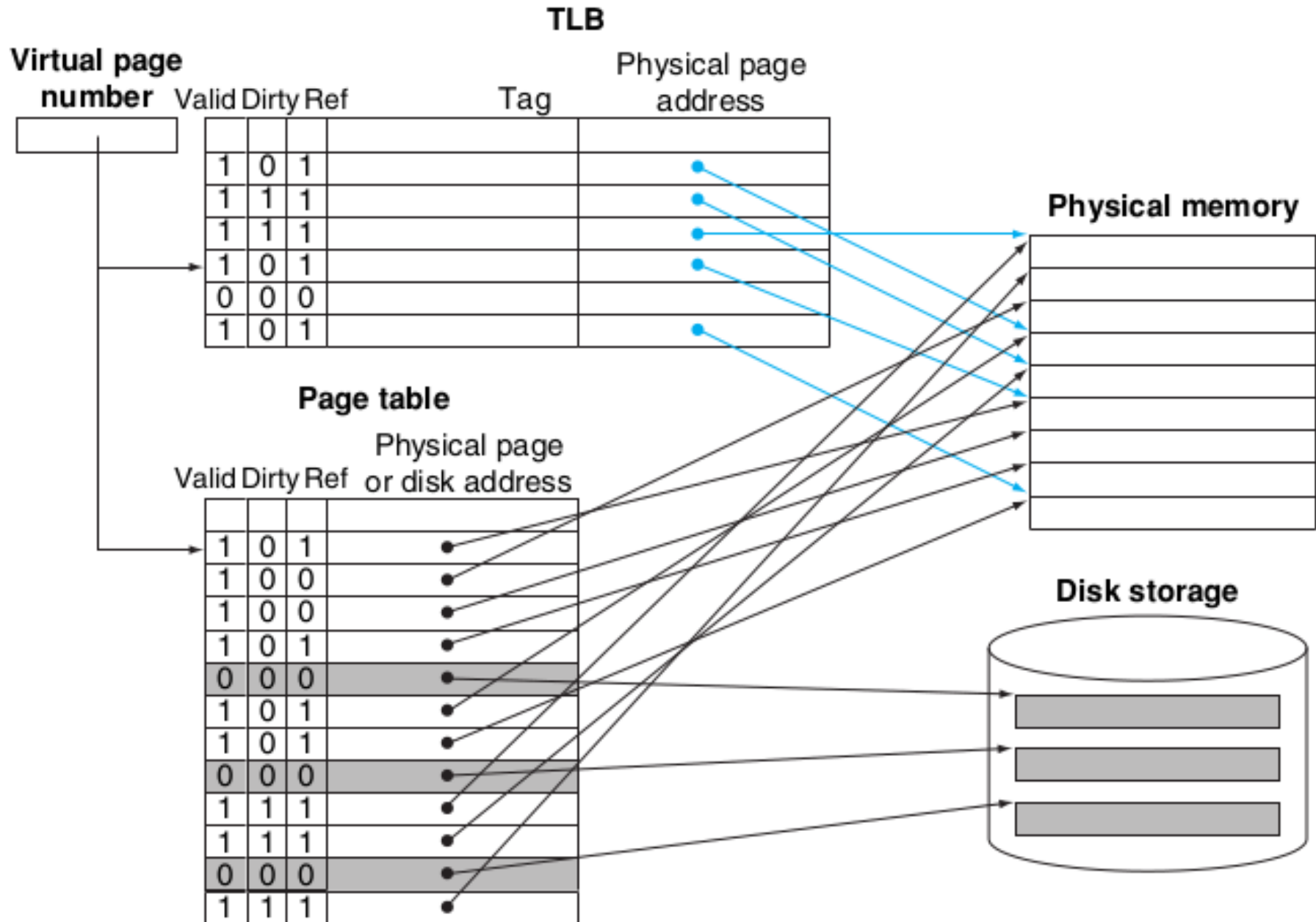


Figure 3-13. Linear Address Translation (4-MByte Pages)

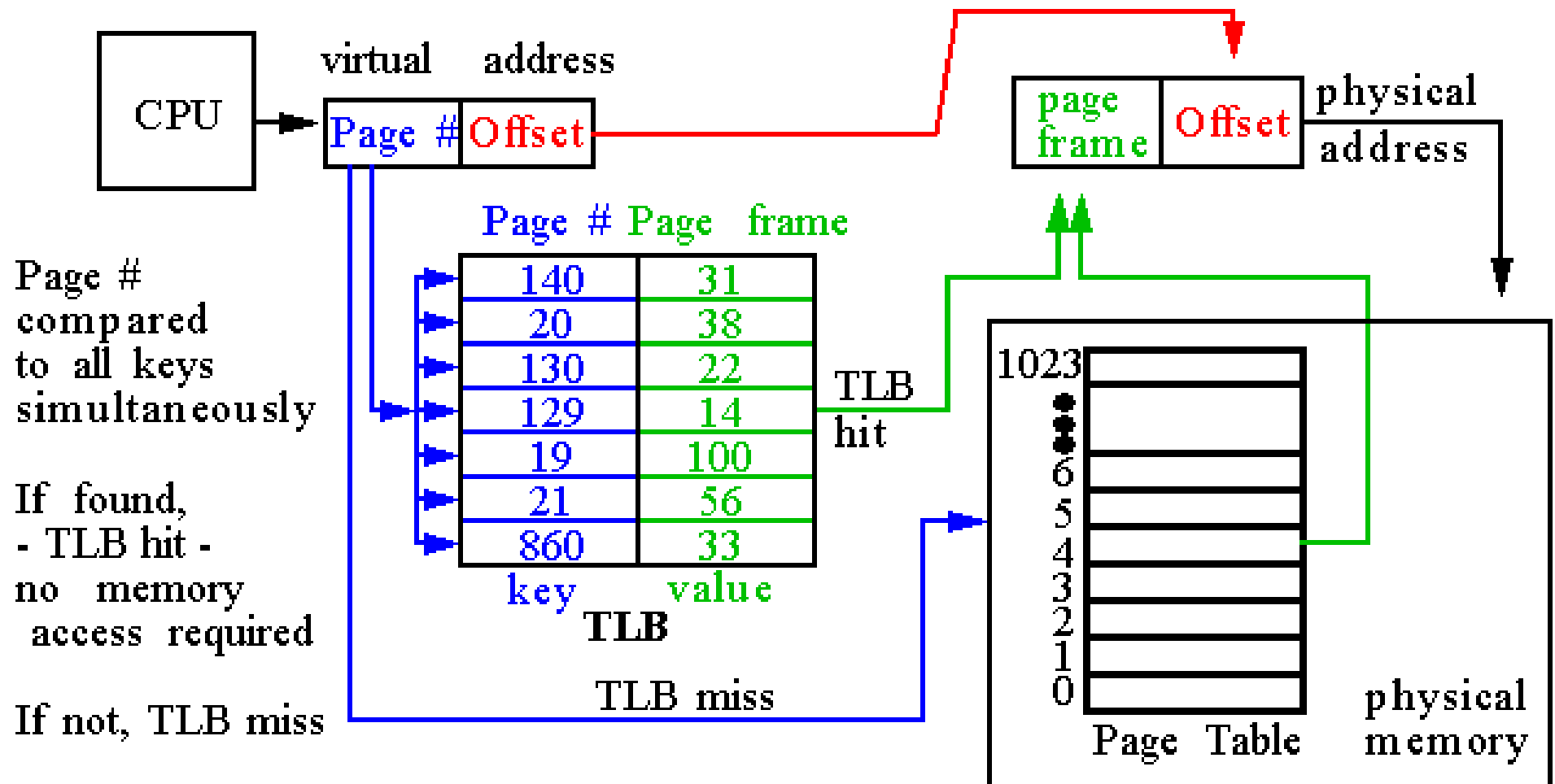
Exemplo



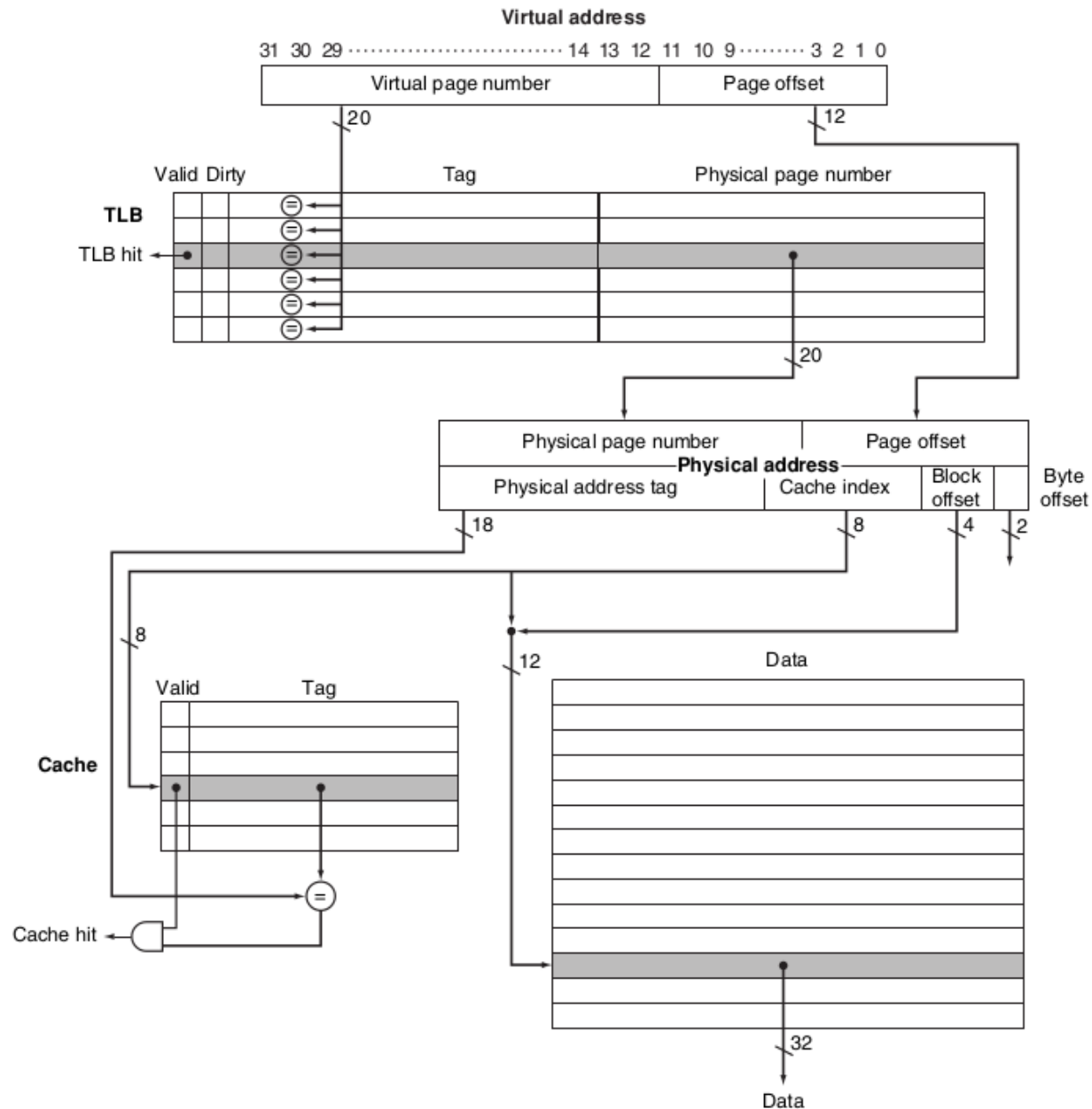
Translation Lookaside Buffer



Exemplo TLB



Exemplo VM (TLB) + CM



Créditos

Figuras:

- Patterson&Hennessy 2018, 2004, Morgan Kaufmann
- Jim Plusquellic, Novo Mexico University



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